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Curriculum Vitae
Gregory C. Chang, PhD, MPH

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PROFESSIONAL SUMMARY

Health economist and epidemiologist (PhD) with expertise in real-world evidence (RWE), health economics and outcomes research (HEOR), pharmacoepidemiology, and pharmacovigilance. Currently generates claims-based real-world evidence for cardiac medical devices and builds business-intelligence dashboards at ZOLL Cardiac Management Solutions. Experienced in integrating clinical trial and real-world datasets across cardiology, oncology, hepatology, gastroenterology, and infectious diseases. Skilled in translating complex data into actionable insights for decision-making, with additional experience in HIV surveillance and global health. Proficient in R, SAS, SQL/Snowflake, Power BI, and AI/LLM validation for evidence synthesis.

EDUCATION

The Ohio State University – PhD, Epidemiology, GPA 3.90. May 2026
Dissertation: Structural and spatial factors associated with HIV viral load suppression in East Africa: evidence from population-based surveys
Columbia University Mailman School of Public Health – MPH, Epidemiology. 2018
Case Western Reserve University – BA, Biology & Anthropology 2016

Key Skills:
Data Analysis: R, R Markdown, SAS, Stata, SQL/Snowflake, Power BI, ArcGIS Pro, QGIS, REDCap
Epidemiology Methods: Real-world evidence, claims data analysis, retrospective cohort analysis, complex survey analysis

PROFESSIONAL EXPERIENCE

ZOLL Cardiac Management Solutions – Health Economics Apr. 2026 – Present
Health Economics Specialist, Columbus, OH

- Lead **real-world evidence analyses** of ZOLL cardiac medical devices – the LifeVest wearable cardioverter defibrillator, the Heart Failure Management System (HFMS), and TherOx SuperSaturated Oxygen (SSO2) Therapy – using large-scale **healthcare claims data**.
- Design **retrospective cohort studies** and statistical analysis plans evaluating device utilization, clinical outcomes, and healthcare resource use to support market access and payer strategy.
- Build **Power BI dashboards** that translate internal operational and commercial data into **business intelligence** for leadership and cross-functional teams.

Analysis Group – Health Economics and Outcomes Research Jun. 2025 – Aug. 2025
Associate Intern, New York, NY

- Executed RWE strategies to support market access and regulatory goals for oncology and immunology assets; integrated clinical trial and real-world data (Komodo Health) to define treatment patterns and healthcare utilization.
- Designed **feasibility assessments** and data collection tools to streamline **chart review studies**, reducing reliance on external vendors and improving efficiency.
- Synthesized evidence on inflammatory bowel disease treatments through an **AI/LLM**-targeted literature review, generating insights into treatment efficacy, safety, and outcomes to inform pharmacoepidemiologic and market access strategies.
- Led **retrospective observational cohort analyses** in relapsed/refractory multiple myeloma to evaluate real-world effectiveness and safety of newly approved bispecific antibody therapies, including adverse events, treatment patterns, and discontinuation outcomes.
- Performed **pharmacovigilance-focused analyses** (e.g., adverse events, time to discontinuation, progression-free survival, and overall survival) in oncology populations and delivered client-ready results through R Markdown visualizations.

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OSUCCC – James, Center for Cancer Health Equity Aug. 2023 – Dec. 2024
Susan G. Komen Training to Eliminate Disparities (TREND) Trainee, Columbus, OH

- Conducted research on breast cancer disparities among minority populations; led analyses on screening adherence before, during, and after the COVID-19 pandemic.
- Applied advanced statistical methods to assess the influence of socioeconomic and geographic factors on screening outcomes.

Nationwide Children's Hospital Center for Injury Research and Policy May 2022 – Jan. 2024
Project Manager, R01, Columbus, OH

- Managed NIH-funded study testing mobile technology interventions to reduce distracted driving among young adults.
- Supervised a 6-member multidisciplinary team, streamlined communication using REDCap and Microsoft Teams.

Centers for Disease Control and Prevention – PHI/CDC Global Health Fellowship Aug. 2018 – Jan. 2022
Epidemiologist, Division of Global HIV & TB, Atlanta, GA

- Supported national HIV surveys (PHIA) in Kenya, Nigeria, Lesotho, Uganda, and Zimbabwe; analyzed complex survey data to validate indicators for final reports.
- Collaborated with Ministries of Health and international partners to implement surveys of 25,000+ participants, using QGIS to optimize deployment plans.

Centers for Disease Control and Prevention COVID-19 Task Force Jun. 2020 – Mar. 2021
Analyst, SARS-CoV-2 Reinfection and Vaccine Breakthrough Unit, Atlanta, GA

- Managed clinician-reported SARS-CoV-2 reinfection investigation, developing REDCap system and specimen handling SOPs.
- Data manager for vaccine breakthrough investigations; built REDCap system for national reporting of cases.

ICAP Mailman School of Public Health, Columbia University May 2017 – May 2018
Research Intern/Analyst, Population-based HIV Impact Assessment (PHIA), Yaoundé, Cameroon

- Supported implementation of a national HIV survey by providing real-time technical assistance to 30 field teams, troubleshooting data collection software (Open Data Kit), and ensuring smooth IT operations.
- Cleaned and validated household, biomarker, and behavioral data, and developed SOPs for data management to strengthen survey data quality and reporting.

Case Western Reserve University Jan. 2015 – May 2016
Research Assistant, Center for Global Health and Disease, Cleveland, OH

- Conducted molecular research on Schistosomiasis, a neglected tropical disease affecting ~240M people annually.
- Performed RT-PCR, yeast cultures, RNA sequencing, schistosome parasite, and DNA extraction.

ARTICLES AND PUBLICATIONS

1. **Chang, Gregory C.**, Bingham, Trista, et al. (2025) Undetectable HIV-1 viral load among virally suppressed people living with HIV: implications for Undetectable equals Untransmittable (U=U) in resource-limited settings using dried spot testing. *JAIDS Journal of Acquired Immune Deficiency Syndromes*. DOI: 10.1097/QAI.0000000000003745
2. **Chang, G. C.**, Ramesh, B., Liebhard, B., Borrasca, W. E., Loux, T., Horan, J., & Turner, A. N. (2024) Census tract-level social vulnerability and gonorrhea rates in central Ohio: results from the STD Surveillance Network, 2020-2021. *Sexually Transmitted Diseases*. DOI: 10.1097/OLQ.0000000000002062
3. Zhang, F., Benedetti, M., **Chang, G.**, Evans, A., Pan, Y., Shoots-Reinhard, B., & Zhu, M. (2024) Demographic characteristics and trends of cell phone use while driving citations in selected states in the United States, 2010–2020. *Traffic Injury Prevention*, 1–7. DOI: 10.1080/15389588.2024.2351605
4. Alexander Evans, Marco H. Benedetti, **Gregory Chang**, et al. (2024) A comparison of licensed driver counts from the Federal Highway Administration and driver licensing agencies in 11 states, *Traffic Injury Prevention*, 25:2, 133-137, DOI: 10.1080/15389588.2023.2278412

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5. Ewing, A. P., **Chang, G. C.**, et al. (2023) Lifestyle Related Cancer Risk and Protective Behaviors Vary among a Convenient Sample of Physically Active, Young-to-Middle-Aged Adults 18-49. *International journal of environmental research and public health*, 20(13), 6305. DOI: 10.3390/ijerph20136305
6. **Chang, G. C.**, West, C. A., et al. (2022) Hazardous alcohol use and HIV indicators in six African countries: results from the Population-based HIV Impact Assessments, 2015-2017. *Journal of the International AIDS Society*, 25(11), e26029. DOI: 10.1002/jia2.26029
7. Achia, T., Cervantes, I. F., Stupp, P., Musingila, P., Muthusi, J., Waruru, A., Schmitz, M., Bronson, M., **Chang, G.**, et al. (2022) Methods for conducting trends analysis: roadmap for comparing outcomes from three national HIV Population-based household surveys in Kenya (2007, 2012, and 2018). *BMC Public Health*, 22(1), 1337. DOI: 10.1186/s12889-022-13633-8
8. West, CA., **Chang, GC**, et al. (2021) Unawareness of HIV Infection Among Men Aged 15-59 Years in 13 Sub-Saharan African Countries: Findings from the Population-Based HIV Impact Assessments, 2015-2019. *JAIDS* AUG 1;87(Suppl 1): S97-S106. DOI: 10.1097/QAI.0000000000002708
9. COVID-19 Vaccine Breakthrough Infections Reported to CDC — United States, January 1–April 30, 2021. (2021) *MMWR Morb Mortal Wkly Rep*. DOI: 10.15585/mmwr.mm7021e3
10. Lee JT, Hesse EM, Paulin HN, Datta D, Katz LS, Talwar A, **Chang G**, et al. (2021) Clinical and Laboratory Findings in Patients with Potential Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) Reinfection, May-July 2020. *Clin Infect Dis*. 73(12):2217-2225. DOI: 10.1093/cid/ciab148
11. Zheng L., Jones C., Ejigu G., George N., Geller A., **Chang GC**, et al. (2021) Countries with delayed COVID-19 introduction – characteristics, drivers, gaps, and opportunities. *Global Health* 17, 28. 10.1186/s12992-021-00678-4

CONFERENCE PRESENTATIONS

1. **Chang GC**, Fisher, JL, Paskett, ED. Disparities in mammography screening adherence throughout phases of the COVID-19 Pandemic: Results from the 2018, 2020, and 2022 Behavioral Risk Factor Surveillance System. 17th AACR Conference on The Science of Cancer Health Disparities In Racial/Ethnic Minorities And The Medically Underserved, Los Angeles, CA, September 2024
2. **Chang GC**, Ramesh, B., Liebhard, B., Turner, AN. Census tract-level social vulnerability and gonorrhea in Ohio: results from the STD Surveillance Network 2020-2021. The International Society for Sexually Transmitted Diseases Research, Chicago, IL, July 2023.
3. **Chang GC.**, O'Connell C., et al. The association between alcoholic substance abuse on key HIV outcomes: Results from six population-based HIV impact assessments completed in sub-Saharan Africa. 14th INTEREST 2020 on HIV Treatment, Pathogenesis, and Prevention, Windhoek, Namibia. Oral abstract. November 2020.
4. **Chang GC.**, Bingham T., et al. "Can Undetectable equals Untransmittable (U=U)" be applied in resource-limited settings: Results of the Population-based HIV Impact Assessment surveys in eleven African countries, 2015-2018. Poster (1082) presented at the Conference on Retroviruses and Opportunistic Infections (CROI) 2020, Boston, Massachusetts, March 2020. CROI 2020 Abstracts

PUBLIC HEALTH AND TECHNICAL REPORTS

1. Ohio Commission on Minority Health. (2025, May 16). *Medical Expert Panel on Preventable Cancers: White Paper – Achieving equity and eliminating preventable cancer disparities within racial and ethnic populations* (Vol. 3). Ohio Commission on Minority Health. Minority Health White Paper
2. **Zimbabwe Population-based HIV Impact Assessment 2020 Final Report.** Harare, Zimbabwe. Provided technical assistance in interview and tablet training, monitored field practice before survey implementation, and data analysis for the survey final report. ZIMPHIA Final Report
3. **National AIDS and STI Control Programme (NASCOP), KENPHIA 2018 Final Report.** Nairobi, Kenya. Provided technical assistance to survey implementation, return of viral load results evaluation, and data analysis for the survey final report. KENPHIA Final Report

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4. **Nigeria HIV/AIDS Indicator Survey 2018 Final Report.** Abuja, Nigeria. Provided technical assistance in field deployment planning using Geographical Information System (GIS) software. NAIIS Final Report

LEADERSHIP AND SERVICE

The Ohio State University – James Cancer Hospital Jan. 2022 – Present
Board Member, James Next-Generation Ambassadors (Survivorship Committee)

- Provide leadership on survivorship initiatives, connecting adolescent and young adult (AYA) survivors with resources and supporting AYA cancer research and fundraising efforts.

PROFESSIONAL SERVICE

Ad hoc peer reviewer for manuscripts submitted to:
2025 - Advances in Drug and Alcohol Research; American Journal of STD
BMC Infectious Disease; BMC Public Health